



Software Engineering Final Lab Documentation

Spring 2026



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2. Project Idea

UniSystem is a desktop-based University Management System developed as a Windows Forms application using C#. The system is designed to centralize and simplify the management of academic activities within a university environment. It provides an Admin Panel through which university staff can monitor key statistics such as total students, available courses, number of professors, and average GPA. The system enables efficient management of student records, course offerings, and grade tracking, all from a single, user-friendly interface. Students and administrators can log in securely, navigate between modules (Dashboard, Students, Courses, and Grades), and perform their respective academic tasks with ease.

3. Problem Statement

Universities often face difficulties in managing academic activities efficiently using traditional or manual systems. Managing student records, course registration, schedules, and grades manually can lead to errors, delays, and data loss. In addition, it becomes difficult for students to access academic information quickly and accurately. The lack of a unified system means that data is scattered across different locations, making it hard for administrators to get a clear overview of university operations. Therefore, there is a clear need for a centralized, reliable system that can organize and manage these processes in a more efficient, accurate, and accessible way.



4. Problem Solution

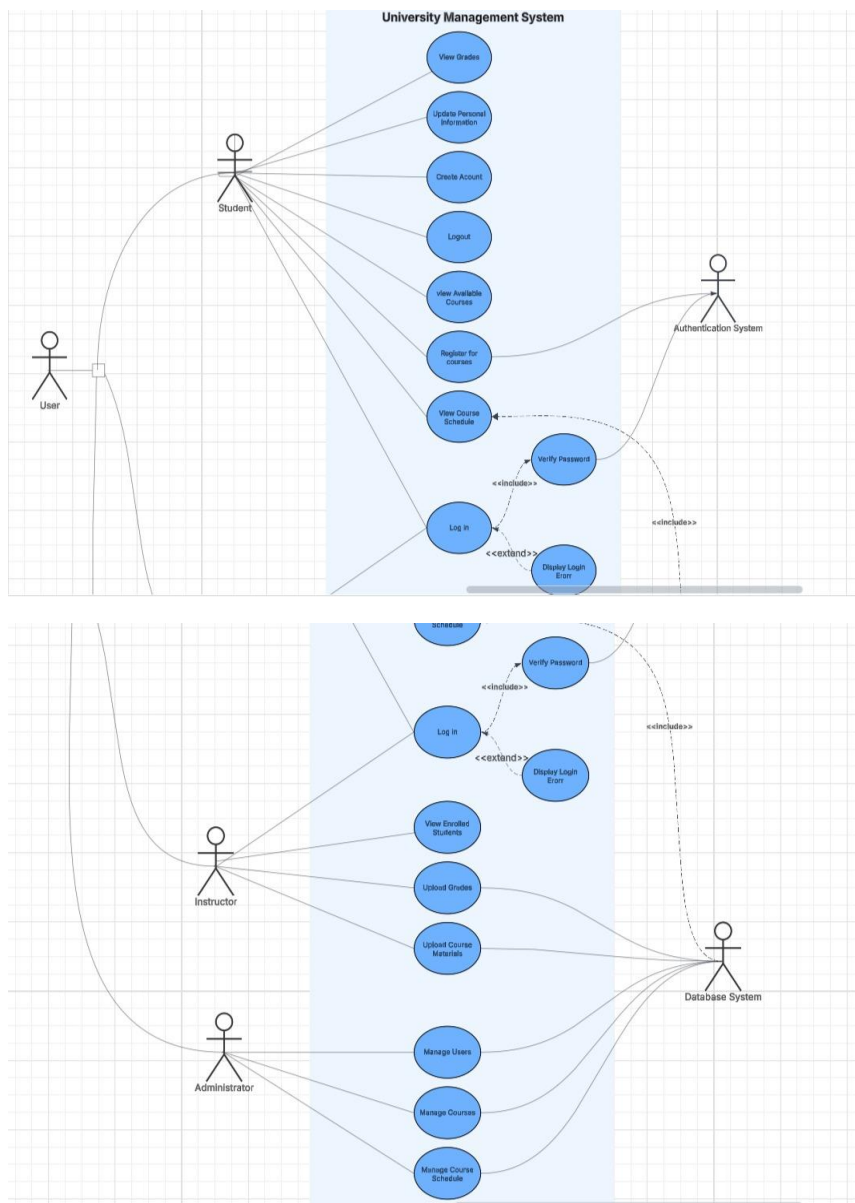
To address these challenges, the UniSystem University Management System is proposed. This system provides a centralized desktop platform that allows students, instructors, and administrators to manage academic activities electronically through an intuitive interface. The system resolves the core problems as follows:

- **Centralized Data Management:** All student records, courses, professors, and grades are stored and managed in one system, eliminating data scatter and inconsistency.
- **Secure Authentication:** The login module enforces credential-based access, ensuring only authorized users can access the system.
- **Real-Time Overview:** The Dashboard displays live statistics including total students (1,248), courses (84), professors (62), and average GPA (3.2), giving administrators instant insights.
- **Efficient Course Management:** Administrators can add or delete courses, view enrollment numbers per course, and monitor which professor teaches each subject.
- **Automated Grade Tracking:** The Grades module automatically calculates totals and assigns letter grades based on midterm and final exam scores, reducing manual calculation errors.
- **Student Record Management:** The Students module provides a searchable list of all students with their ID, department, year, GPA, email, and enrollment status.

5. System Analysis

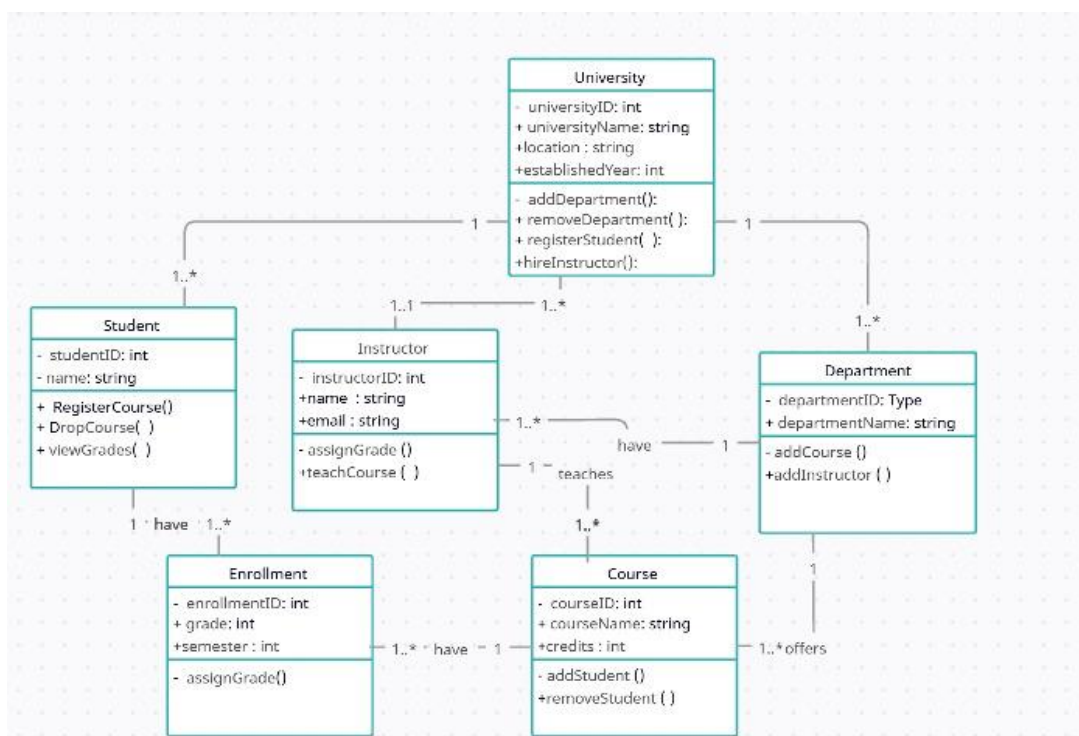
5.1 Use Case Diagram

The Use Case Diagram illustrates the interactions between system actors (Student, User, Instructor, Administrator, Authentication System, and Database System) and the available system functions. The Student actor can create an account, log in, log out, view available courses, register for courses, view course schedule, view grades, and update personal information. The Instructor can view enrolled students, upload grades, and upload course materials. The Administrator manages users, courses, and course schedules. The Authentication System handles login verification, with "include" relationships for verifying passwords and "extend" relationships for displaying login errors.



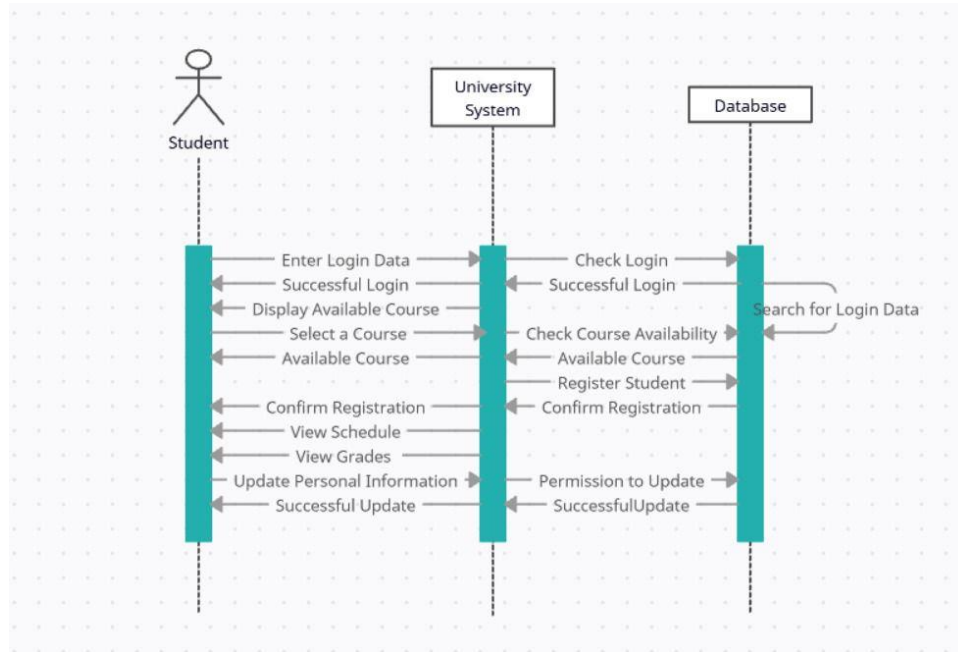
5.2 Class Diagram

The Class Diagram shows the main entities of the system and their relationships. The central University class holds universityID, universityName, location, and establishedYear, with methods to addDepartment, removeDepartment, registerStudent, and hireInstructor. It is associated with Student (studentID, name; RegisterCourse, DropCourse, viewGrades), Instructor (instructorID, name, email; assignGrade, teachCourse), and Department (departmentID, departmentName; addCourse, addInstructor). The Course class (courseID, courseName, credits; addStudent, removeStudent) is connected to Department and Instructor. The Enrollment class (enrollmentID, grade, semester; assignGrade) links Student to Course in a 1-to-many relationship.



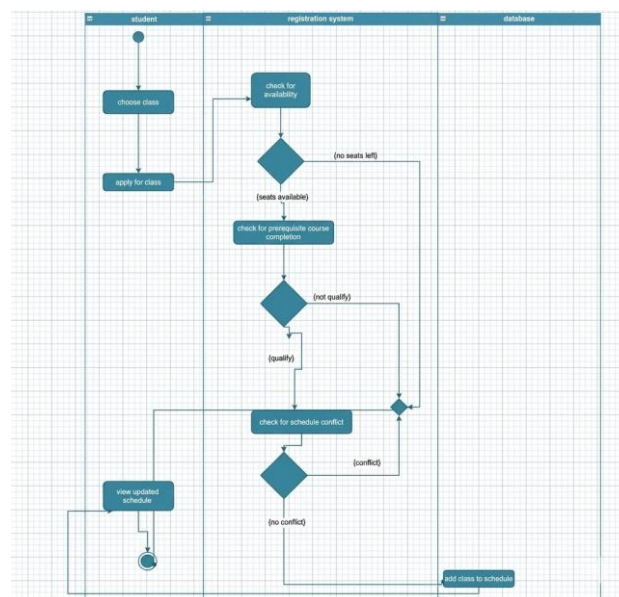
5.3 Sequence Diagram

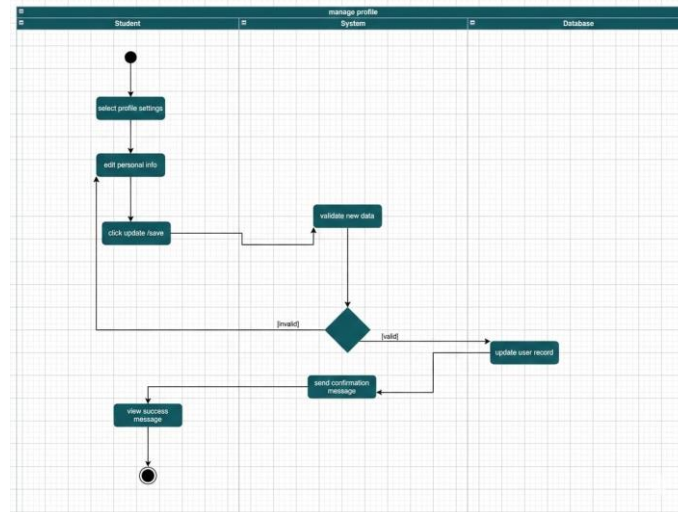
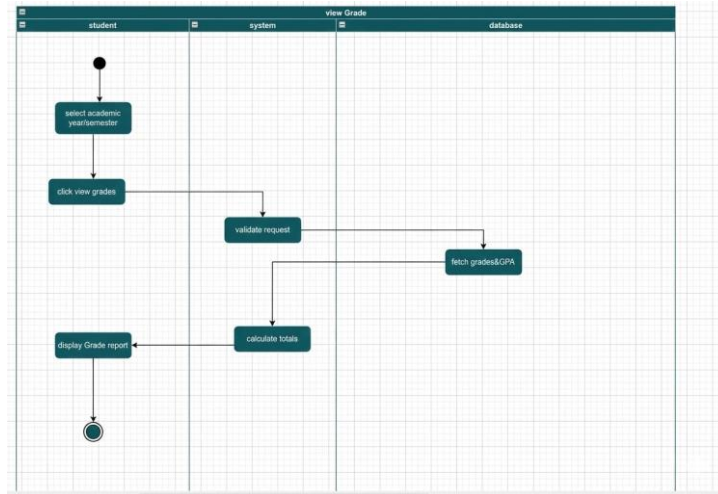
The Sequence Diagram depicts the flow of messages between the Student, the University System, and the Database across the main use cases. The student enters login credentials; the system sends a Check Login request to the database which searches for the login data and returns a Successful Login. The system then displays available courses. The student selects a course; the system checks availability with the database, confirms registration, and the student can then view their schedule and grades. When the student updates personal information, the system requests permission from the database and returns a Successful Update confirmation.



5.4 Activity Diagram

The three activity diagrams cover: (1) Course Registration — choosing a class, checking seat availability, verifying prerequisites, checking schedule conflicts, then adding to schedule; (2) View Grades — selecting academic year/semester, clicking View Grades, the system fetching and calculating totals, then displaying the grade report; (3) Manage Profile — selecting profile settings, editing personal info, validating the new data, updating the user record, and confirming success.



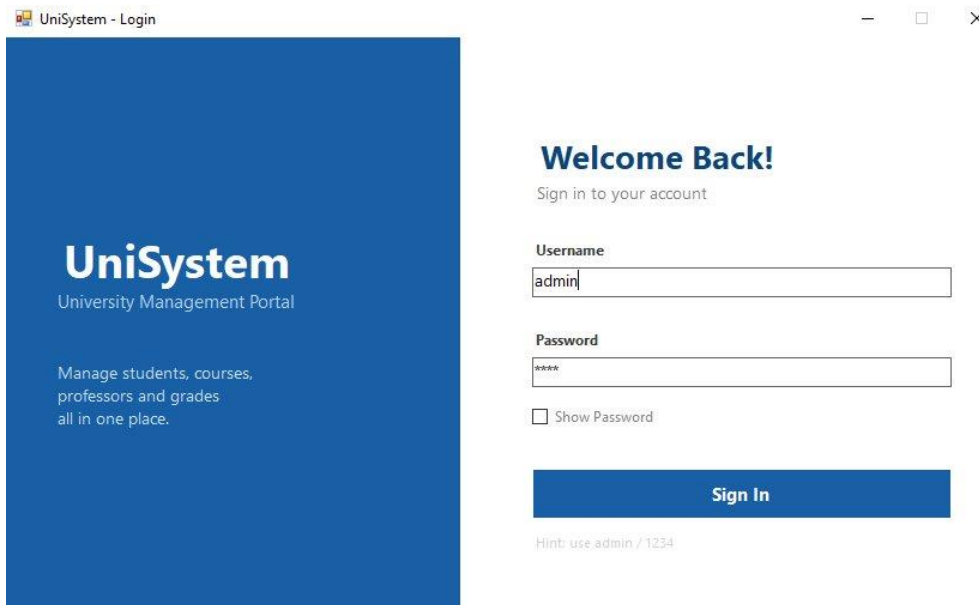


6. Implementation

The following screenshots show the implemented screens and the corresponding C# WinForms source code of the UniSystem application.

Login Screen

The login screen prompts the user for a username and password. A "Show Password" checkbox is available. The hint text indicates the default credentials (admin / 1234). On successful login, the user is redirected to the Dashboard.



The screenshot shows the UniSystem - Login window. On the left, a blue panel displays the UniSystem logo and the text "University Management Portal" and "Manage students, courses, professors and grades all in one place." On the right, the login form includes a "Welcome Back!" greeting, a "Sign in to your account" prompt, and fields for "Username" (containing "admin") and "Password" (masked with asterisks). A "Show Password" checkbox is present. A blue "Sign In" button is at the bottom, with a hint "Hint: use admin / 1234" below it.

Dashboard

The Dashboard provides an overview of university statistics: Total Students (1,248), Courses (84), Professors (62), and Average GPA (3.2). Below the stats, a "Recent Enrollments" table lists the latest student registrations with course code, department, date, and status.



```

1  using System;
2  using System.Drawing;
3  using System.Windows.Forms;
4
5  namespace UniversitySystem
6  {
7      4 references
8      public class LoginForm : Form
9      {
10         private Panel leftPanel;
11         private Panel rightPanel;
12         private Label lblTitle;
13         private Label lblSubtitle;
14         private Label lblWelcome;
15         private Label lblDesc;
16         private Label lblUsername;
17         private Label lblPassword;
18         private TextBox txtUsername;
19         private TextBox txtPassword;
20         private Button btnLogin;
21         private Label lblError;
22         private CheckBox chkShowPassword;
23
24         2 references
25         public LoginForm()
26         {
27             InitializeComponent();
28         }
29
30         1 reference
31         private void InitializeComponent()
32         {
33             this.Text = "UniSystem - Login";
34             this.Size = new Size(850, 520);
35             this.StartPosition = FormStartPosition.CenterScreen;
36             this.FormBorderStyle = FormBorderStyle.FixedSingle;
37             this.MaximizeBox = false;
38             this.BackColor = Color.White;
39         }
40     }
41 }

```

Source Code – MainForm.cs (Form Settings & Sidebar)

The main form is titled "UniSystem – University Management" (1100×660), with a dark sidebar panel containing the logo, admin label, and navigation buttons for Dashboard, Students, Courses, and Grades. Click handlers switch active views.

```

27 // --- FORM SETTINGS ---
28 this.Text = "UniSystem - University Management";
29 this.Size = new Size(1100, 660);
30 this.StartPosition = FormStartPosition.CenterScreen;
31 this.BackColor = Color.FromArgb(244, 246, 250);
32 this.MinimumSize = new Size(900, 600);
33
34 // --- SIDEBAR ---
35 sidePanel = new Panel();
36 sidePanel.Size = new Size(200, 660);
37 sidePanel.Location = new Point(8, 0);
38 sidePanel.BackColor = Color.FromArgb(12, 68, 134);
39 sidePanel.Anchor = AnchorStyles.Left | AnchorStyles.Top | AnchorStyles.Bottom;
40
41 Label lblLogo = new Label();
42 lblLogo.Text = "UniSystem";
43 lblLogo.Font = new Font("Segoe UI", 15, FontStyle.Bold);
44 lblLogo.ForeColor = Color.White;
45 lblLogo.Location = new Point(28, 24);
46 lblLogo.AutoSize = true;
47
48 Label lblAdmin = new Label();
49 lblAdmin.Text = "Admin Panel";
50 lblAdmin.Font = new Font("Segoe UI", 9);
51 lblAdmin.ForeColor = Color.FromArgb(150, 190, 230);
52 lblAdmin.Location = new Point(28, 50);
53 lblAdmin.AutoSize = true;
54
55 Panel divider = new Panel();
56 divider.Size = new Size(160, 1);
57 divider.Location = new Point(28, 72);
58 divider.BackColor = Color.FromArgb(50, 100, 160);
59
60 sidePanel.Controls.Add(lblLogo);
61 sidePanel.Controls.Add(lblAdmin);
62 sidePanel.Controls.Add(divider);
63
64 // Nav buttons
65 btnDashboard = CreateNavButton("Dashboard", 90, "\ud83d\udd12");
66 btnStudents = CreateNavButton("Students", 130, "\ud83d\udd65");
67 btnCourses = CreateNavButton("Courses", 170, "\ud83d\uddcd");
68 btnGrades = CreateNavButton("Grades", 210, "\ud83d\uddcc");
69
70 btnDashboard.Click += (s, e) => { SetActive(btnDashboard); ShowDashboard(); };
71 btnStudents.Click += (s, e) => { SetActive(btnStudents); ShowStudents(); };
72 btnCourses.Click += (s, e) => { SetActive(btnCourses); ShowCourses(); };
73 btnGrades.Click += (s, e) => { SetActive(btnGrades); ShowGrades(); };
74
75 sidePanel.Controls.Add(btnDashboard);
76 sidePanel.Controls.Add(btnStudents);
77 sidePanel.Controls.Add(btnCourses);
78 sidePanel.Controls.Add(btnGrades);

```

Source Code – Program.cs

The application entry point uses the [STAThread] attribute, enables visual styles, and launches the LoginForm as the startup window via Application.Run.


```

1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Threading.Tasks;
5  using System.Windows.Forms;
6
7  namespace UniversitySystem
8  {
9      internal static class Program
10     {
11
12         [STAThread]
13         static void Main()
14         {
15             Application.EnableVisualStyles();
16             Application.SetCompatibleTextRenderingDefault(false);
17             Application.Run(new LoginForm());
18         }
19     }
20 }
21

```

Source Code – AssemblyInfo.cs

The assembly metadata file defines the application title as "UniversitySystem", sets the copyright to 2026, and specifies assembly version 1.0.0.0.

```

1  using System.Reflection;
2  using System.Runtime.CompilerServices;
3  using System.Runtime.InteropServices;
4
5
6  [assembly: AssemblyTitle("UniversitySystem")]
7  [assembly: AssemblyDescription("")]
8  [assembly: AssemblyConfiguration("")]
9  [assembly: AssemblyCompany("")]
10 [assembly: AssemblyProduct("UniversitySystem")]
11 [assembly: AssemblyCopyright("Copyright © 2026")]
12 [assembly: AssemblyTrademark("")]
13 [assembly: AssemblyCulture("")]
14
15 [assembly: ComVisible(false)]
16
17 [assembly: Guid("664a41b5-a2c3-4ee2-9193-e89f9f24b7b2")]
18
19
20 [assembly: AssemblyVersion("1.0.0.0")]
21
22
23 [assembly: AssemblyFileVersion("1.0.0.0")]
24
25

```

Source Code – Resources.cs

The auto-generated Resources class manages strongly-typed resource access using a ResourceManager for "UniversitySystem.Properties.Resources", supporting globalization via CultureInfo.

```

namespace UniversitySystem.Properties
{
    [global::System.CodeDom.Compiler.GeneratedCodeAttribute("System.Resources.Tools.StronglyTypedResourceBuilder", "4.8.0.0")]
    [global::System.Diagnostics.DebuggerNonUserCodeAttribute()]
    [global::System.Runtime.CompilerServices.CompilerGeneratedAttribute()]
    internal class Resources
    {
        private static global::System.Resources.ResourceManager resourceMan;
        private static global::System.Globalization.CultureInfo resourceCulture;

        [global::System.Diagnostics.CodeAnalysis.SuppressMessageAttribute("Microsoft.Performance", "CA1811:AvoidUncalledPrivateCode")]
        internal Resources()
        {
        }

        [global::System.ComponentModel.EditorBrowsableAttribute(global::System.ComponentModel.EditorBrowsableState.Advanced)]
        internal static global::System.Resources.ResourceManager ResourceManager
        {
            get
            {
                if ((resourceMan == null))
                {
                    global::System.Resources.ResourceManager temp = new global::System.Resources.ResourceManager("UniversitySystem.Properties.resources", typeof(Resources).Assembly);
                    resourceMan = temp;
                }
                return resourceMan;
            }
        }

        [global::System.ComponentModel.EditorBrowsableAttribute(global::System.ComponentModel.EditorBrowsableState.Advanced)]
        internal static global::System.Globalization.CultureInfo Culture
        {
            get
            {
                return resourceCulture;
            }
            set
            {
                resourceCulture = value;
            }
        }
    }
}

```

Source Code – Settings.cs

The Settings class is a sealed partial class inheriting from ApplicationSettingsBase, providing a synchronized singleton instance via the Default property for application-level configuration management.

```

1 namespace UniversitySystem.Properties
2 {
3
4
5
6     [global::System.Runtime.CompilerServices.CompilerGeneratedAttribute()]
7     [global::System.CodeDom.Compiler.GeneratedCodeAttribute("Microsoft.VisualStudio.Editors.SettingsDesigner.SettingsSingleFileGenerator", "11.0.0.0")]
8     internal sealed partial class Settings : global::System.Configuration.ApplicationSettingsBase
9     {
10
11         private static Settings defaultInstance = ((Settings)(global::System.Configuration.ApplicationSettingsBase.Synchronized(new Settings())));
12
13         [global::System.ComponentModel.EditorBrowsableAttribute(global::System.ComponentModel.EditorBrowsableState.Advanced)]
14         public static Settings Default
15         {
16             get
17             {
18                 return defaultInstance;
19             }
20         }
21     }
22 }

```


7. Testing

The following test cases were executed to verify the correctness and completeness of the UniSystem application. All test cases passed successfully.

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01: Login – Valid Credentials	Username: admin, Password: 1234	Login succeeds; Dashboard loads	Dashboard displayed	Pass
TC-02: Login – Invalid Credentials	Username: admin, Password: wrong	Error message shown; no navigation	Error message displayed	Pass
TC-03: View Students List	Navigate to Students page	All student records displayed in table	Students table shown	Pass
TC-04: Search Students	Type name in search box	List filters to matching students	Filter works correctly	Pass
TC-05: View Courses	Navigate to Courses page	All courses with details displayed	Courses table shown	Pass
TC-06: Add Course	Click "+ Add Course" button	Course form opens for input	Form opens	Pass
TC-07: Delete Course	Select course, click "Delete"	Course removed from list	Course removed	Pass
TC-08: View Grades	Navigate to Grades page	Grade table with Midterm/Final/Total shown	Grade table displayed	Pass
TC-09: Filter Grades by Semester	Select "Spring 2026" from dropdown	Only Spring 2026 grades shown	Filter applied	Pass
TC-10: View Dashboard Statistics	Log in and view Dashboard	Total students, courses, GPA displayed	Stats displayed	Pass
TC-11: Logout	Click "Logout" in sidebar	Session ends; login screen shown	Redirected to login	Pass
TC-12: View Recent Enrollments	Open Dashboard	Recent enrollment table displayed	Table shown	Pass

All 12 test cases passed. The system correctly handles both valid and invalid inputs, performs all CRUD operations on courses, displays and filters student/grade data, and manages user sessions.